Technical Supplement to “A Two-Thirds Hyperbolicity Wedge for Jensen Polynomials of Riemann’s Xi-Function”

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Abstract

This supplement records the quantitative inequalities, normalization audits, formal theorem map, exact-computation protocols, and reproducibility boundary for the accompanying main paper. It is intended to make every compressed step inspectable without searching historical project notes. Numerical enclosures are clearly separated from uniform analytic proofs. All review evidence currently available is AI review, not human or peer review.

Contents

1 Notation and fixed domains

The centered coefficient convention is

$$\xi(1/2 + w) = \sum_{n \geq 0} \gamma(n) w^{2n}/n!.$$

The continued coefficient variable is M, the leading Mellin index is $N = 2M - 2$, and the saddle satisfies

$$N = L_N(\pi e^{L_N} + 3/4).$$

We use

$$r = L_N^{-1}, \qquad \sigma = L_N/N, \qquad Q_N = (1 + L_N)N - 3L_N^2/4, \qquad Q_N = L_N^2 K_N.$$

The final nested sectors are

$$\mathcal{S}_1(R) = \{M : |M| > R,\ |\arg M| < 1/200\},$$
$$\overline{\mathcal{S}}_0(R) = \{M : |M| \geq R,\ |\arg M| \leq 1/400\}.$$

All complex logarithms, powers, and square roots are obtained by continuation from the positive real axis in $\mathcal{S}_1(R)$. Constants in a statement labeled uniform are uniform only after these sectors and the parameter box have been fixed.

The parameter variables and their order are

$$(\alpha, t, w, \delta), \qquad y_* = (3, 2, 16/3, 1/3).$$

The hypergeometric scale is always $D/(AC)$, and the finite-free root order is $\lambda_1 \geq \cdots \geq \lambda_d > 0$.

2 Exact completed-xi and Mellin normalization

Mathlib defines the completed zeta $\Lambda(s)$ and its pole-removed entire version $\Lambda_0(s)$ with

$$\Lambda(s) = \Lambda_0(s) - \frac{1}{s} - \frac{1}{1-s}.$$

The repository's entire xi is

$$\xi(s) = \frac{s(s-1)}{2}\Lambda_0(s) + \frac{1}{2}. \tag{2.1}$$

Away from $0, 1$, elementary algebra converts (2.1) to the usual $s(s-1)\Lambda(s)/2$. Mathlib's functional equation for Λ_0 implies $\xi(1-s) = \xi(s)$, hence evenness at $1/2$.

The Lean definition

$$\gamma(n) = \frac{n!}{(2n)!} D^{2n}[w \mapsto \xi(1/2 + w)](0)$$

is therefore exactly the paper convention. The concrete integral seam is

$$D^{2n}[w \mapsto \xi(1/2 + w)](0) = 8 \int_0^\infty \omega(e^{2u}) e^{u/2} u^{2n}\ \mathrm{d}u. \tag{2.2}$$

Lean proves that (2.2) implies the paper's factor-eight formula. It does not presently prove (2.2) for the concrete theta kernel; that identity is proved on paper from Riemann's integral.

For source comparison, define

$$F(s) = \int_0^\infty \mathcal{K}(t) t^s\ \mathrm{d}t$$

with the exact kernel $\mathcal{K}$ fixed by the theta-mode expansion. Direct integration by parts gives

$$\gamma(M) = \frac{\Gamma(M+1)}{\Gamma(2M+1)} 2^{-2M-2} \left(32 \binom{2M}{2} F(2M-2) - F(2M) \right). \tag{2.3}$$

The factors in (2.3) pass three independent checks: completed-zeta Taylor balls, the positive omega integral, and separate integration of the two F moments with analytic tails. Define

$$M_z = 2^{-2z-2} \left(32 \binom{2z}{2} F(2z-2) - F(2z) \right), \qquad h(z) = \log M_z,$$

on the remote sectorial branch real on the positive axis. Thus $\gamma(z) = \Gamma(z+1)M_z/\Gamma(2z+1)$ exactly. The derivative tower below is the tower of $\log M_z$, not of $\log \gamma(z)$.

3 Rouché bounds for the saddle

Let $\ell = \log |N| \geq 12$ and

$$L_0 = \log N - \log\log N - \log \pi.$$

On $|L - L_0| \leq 1$ set

$$m(\ell) = \ell - \log(\ell+1) - \log \pi - 1,$$
$$M_\theta(\ell) = \ell + 2 + \log(\ell+1) + \theta/\ell + \log \pi.$$

The elementary estimates used in the main proof are

$$\Re L \geq m(\ell) > 0, \qquad |L| \leq M_\theta(\ell), \tag{3.1}$$

$$\left| \frac{3L}{4N} \right| \leq \frac{3M_\theta(\ell)}{4e^\ell} < 10^{-4}, \tag{3.2}$$

$$\left| \log\left(1 - \frac{3L}{4N}\right) \right| \leq \frac{3M_\theta(\ell)}{2e^\ell}. \tag{3.3}$$

For $|z| \leq 1/2$, the segment integral $\log(1+z) = \int_0^1 z/(1+tz)\, dt$ gives $|\log(1+z)| \leq 2|z|$. Apply this to the difference between $\log L$ and $\log L_0$, using $|L_0| \geq \ell/2$, to obtain

$$\sup_{|L-L_0|=1} |G_N(L) - (L - L_0)| \leq 2\frac{\log(\ell+1) + \theta/\ell + \log\pi + 1}{\ell} + \frac{3M_\theta(\ell)}{2e^\ell}. \tag{3.4}$$

The right side decreases after $\ell = 12$ and is below one there. The independent exact boundary audit reports 0.807 for the deliberately wider test angle. Rouché therefore gives one zero in the disc.

The denominator check precedes implicit differentiation. From $|r|, |\sigma| \leq 7/50$,

$$|1 + r - 3\sigma/4| \geq 151/200 > 9/16, \tag{3.5}$$

$$|4 + 4r - 3\sigma| \geq 151/50. \tag{3.6}$$

Thus $Q_N \neq 0$, the root is simple, and the local analytic roots patch by uniqueness. A generic complex closed-disc contraction theorem has also been formalized, but the paper retains the Rouché route because the latter's boundary estimates are already explicit.

4 Imaginary-part and curvature bootstrap

Write $L = a + ib$ and $N = |N|e^{i\phi}$. The argument of the saddle equation is

$$\phi = \arg L + \arg(\pi e^L + 3/4). \tag{4.1}$$

Since $a \geq m(12) > 7.29$, the second term differs from b by at most $O(e^{-a})$. The first is bounded by $|b|/a$. Hence

$$|b| \leq |\phi| + |b|/a + \frac{3e^{-a}}{2\pi}. \tag{4.2}$$

For $|\phi| \leq 1/100$, solving (4.2) gives $|b| < 0.012 < 1/20$. This avoids the enormous threshold that would result from bounding $|L_N - L_0|$ alone.

Now

$$K_N = \frac{N}{L_N}\left(1 + \frac{1}{L_N} - \frac{3L_N}{4N}\right).$$

The factor in parentheses lies in the disc of radius 49/200 about 1. Combining its argument with $\arg N - \arg L_N$ gives $|\arg K_N| < 1/20$ after increasing R. Therefore

$$\Re K_N \geq \cos(1/20)|K_N|. \tag{4.3}$$

The stronger right-half-plane statement (4.3), not merely $K_N \neq 0$, is what controls the complex Gaussian.

5 Contour, connector, and Gaussian estimates

Set

$$\Phi_N(u) = N \operatorname{Log} u - \frac{3}{4}u - \pi e^u, \qquad g_N(u) = e^u e^{\Phi_N(u)}, \qquad I_1(N) := F_1(N) := \int_0^\infty g_N(u)\ \mathrm{d}u.$$

Write $L_N = \alpha + ib$ and split this integral at 1. Apply Cauchy's theorem to the finite rectangle with vertices $1, X, X + ib, 1 + ib$ and then let $X \to \infty$. The right connector vanishes, and the exact identity is

$$I_1(N) = E_N + \int_1^\infty g_N(x + ib)\ \mathrm{d}x, \qquad E_N = \int_0^1 g_N(x)\ \mathrm{d}x + i\int_0^b g_N(1 + iy)\ \mathrm{d}y. \tag{5.1}$$

The shifted ray lies entirely in $\Re u \geq 1$. Direct comparison with the saddle main term gives

$$|E_N| \leq |\mathcal{A}(N)|e^{-c'|N|\log\log|N|}. \tag{5.2}$$

On the actual shifted ray put $u = L_N + r$, so $r \geq 1 - \Re L_N$, and set $V = |K_N|^{-2/5}$. The central window $|r| \leq V$ is contained in that ray for sufficiently large $|N|$. There,

$$\Phi_N(L_N + r) - \Phi_N(L_N) = -K_N r^2/2 + \kappa_3 r^3/6 + R_4(N, r), \tag{5.3}$$

where

$$|R_4(N, r)| \leq C|K_N||r|^4. \tag{5.4}$$

The Jacobian outside the phase contributes e^r. The Gaussian majorant from (4.3) makes the quartic remainder integrable. The omitted part of the full real Gaussian to the left of $1 - \Re L_N$ is exponentially small, so the full real Gaussian is only a comparison integral. Completing the square gives

$$\frac{\int_{\mathbb{R}} r^3 e^{r - K_N r^2/2}\ \mathrm{d}r}{\int_{\mathbb{R}} e^{r - K_N r^2/2}\ \mathrm{d}r} = \frac{3}{K_N^2} + \frac{1}{K_N^3}.$$

Since $\kappa_3 = O(|K_N|)$, the signed cubic, its square, and (5.4) contribute $O(|K_N|^{-1})$ relatively.

On the actual range $r \geq 1 - \Re L_N$, strict horizontal concavity makes the left part $1 \leq x \leq \Re L_N - V$ increase toward the saddle and gives a negative derivative of size $\gg |K_N|^{3/5}$ at $x = \Re L_N + V$ on the right. Splitting the two resulting tail bounds at $|r| = 1$ makes their Gaussian and exponential integrals elementary. Together with (5.2),

$$F_1(N) = \sqrt{2\pi/K_N}\, L_N^N \exp(L_N/4 - N/L_N + 3/4)\left(1 + O(|K_N|^{-1})\right). \tag{5.5}$$

6 Higher theta modes and sectorial assembly

For mode $m \geq 2$, the phase difference from $m = 1$ contains

$$-\pi(m^2 - 1)e^{L_N + r}. \tag{6.1}$$

Because $|\Im L_N| < 1/20$, its real part on the central window is at most $-c(m^2 - 1)e^{\Re L_N}$. The saddle equation gives $e^{\Re L_N} \asymp |N|/|L_N|$. Thus

$$|F_m(N)| \leq |\mathcal{A}(N)| \exp\left(-c(m^2 - 1)|N|/\log|N|\right). \tag{6.2}$$

Summation over $m \geq 2$ is dominated by the $m = 2$ term.

For the coefficient assembly, write $a(s) = \log \mathcal{A}(s)$. Direct differentiation and the saddle relation give

$$a'(s) = \log L_s + (L_s K_s)^{-1} - \tfrac{1}{2} K_s'/K_s.$$

On a unit step, $a''(s) = O(1/(|s| \log|s|))$, so

$$\frac{F(N+2)}{F(N)} = L_N^2 \left(1 + O\left(\frac{\log|N|}{|N|}\right)\right). \tag{6.3}$$

The sectorial Stirling expansion is applied to each gamma factor with a remainder uniform on the outer fixed sector. Combining (2.3), (5.5), and (6.3) produces the main term in the paper and a holomorphic relative error bounded by $C \log|M|/|M|$.

7 Cauchy transport and derivative artifacts

If f and g are holomorphic on a domain U, g is nonzero, and

$$E = f/g - 1, \qquad |E| \leq \epsilon$$

on a disc of radius R about z_0 contained in U, Cauchy's formula gives

$$|E^{(j)}(z_0)| \leq j!\epsilon/R^j. \tag{7.1}$$

This statement, including the exact factorial and every order through six, is Lean-checked. The analytic input is the uniform sectorial bound, not a finite grid.

The saddle main term is

$$G_0(N) = (N+1)\log L_N + \frac{L_N}{4} - \frac{N}{L_N} - \frac{1}{2}\log Q_N.$$

The exact saddle tower differentiates G_0, using $r = 1/L$ and $\sigma = L/N$. The fifth derivative at $\sigma = 0$ reduces to

$$-\frac{6 + 47r + 146r^2 + 210r^3 + 126r^4 + 42r^5 + 6r^6}{(1+r)^7}. \tag{7.2}$$

At sixth order the reduced denominator is $(4 + 4r - 3\sigma)^{12}$. The numerator has 82 terms and total degree 13. Rather than typeset all 82 monomials, the canonical expanded expression is the output of the exact symbolic producer and the clean-room Mathematica ledger; both are hash-frozen. The coefficientwise box majorant is exactly

$$\frac{6422139805764931584036533551104}{702576099728137594188684005} = 9140.8458\ldots < 10^4.$$

No binary floating-point value is used in this inequality.

8 Four-parameter limiting system

The limiting equations $H^\infty + S^\infty = 0$ are

$$\alpha^{-1} + wt^{-2} + \delta = 2, \tag{8.1}$$
$$wt^{-3} + \delta = 1, \tag{8.2}$$
$$3wt^{-4} + 3\delta = 2, \tag{8.3}$$
$$4wt^{-5} + 4\delta = 2. \tag{8.4}$$

Writing F_j for the left side minus the right side in (8.1)–(8.4), exact subtraction gives

$$3F_1 - F_2 = \frac{3w(t-1)}{t^4} - 1, \qquad \frac{4}{3}F_2 - F_3 = \frac{4w(t-1)}{t^5} - \frac{2}{3}.$$

At a zero these imply $3w(t-1) = t^4$ and $6w(t-1) = t^5$, so positivity gives $t = 2$, followed by $(\alpha, w, \delta) = (3, 16/3, 1/3)$. The exact Jacobian inverse is

$$\begin{pmatrix} -9 & 45 & -24 & 9 \\ 0 & 6 & -6 & 3 \\ 0 & 48 & -112/3 & 16 \\ 0 & 1 & -4/3 & 1 \end{pmatrix}, \tag{8.5}$$

whose largest absolute row sum is 304/3.

The Lean branch interface separates two certificates:

1. a center residual $\|G_n(y_*)\|_\infty \le (3/608)R$;
2. a whole-box derivative defect $\|D(y - PG_n(y))\|_\infty \le 1/2$ for every y in the closed box.

The first implies center displacement at most $R/2$ after multiplication by P; the second supplies both a self-map and contraction. A center-only Jacobian sample is explicitly insufficient.

The outer rational box has exact margins from y_*:

$$(1/2, 1/2), \quad (1/4, 1/4), \quad (1/3, 2/3), \quad (1/12, 1/12).$$

For $e \le 1/12$, interval endpoints prove $A > B > C > D > 0$ and the preliminary Jacobi inequalities without any empirical optimization.

9 Finite-free orientation and localization

For ascending coefficients $p(X) = \sum_{k=0}^{d} (-1)^k a_k X^k$ and $q(X) = \sum_{k=0}^{d} (-1)^k b_k X^k$, define

$$(p \boxtimes_d q)(X) = \sum_{k=0}^{d} (-1)^k \binom{d}{k}^{-1} a_k b_k X^k. \tag{9.1}$$

In monic descending form, the same operation multiplies normalized elementary symmetric coefficients. Fixed-degree reflection satisfies

$$\mathrm{rev}_d(p \boxtimes_d q) = \mathrm{rev}_d(p) \boxtimes_d \mathrm{rev}_d(q).$$

For nonzero constant term, reflection maps roots to reciprocals. Positive interval $[a, b]$ therefore maps to $[b^{-1}, a^{-1}]$.

The transported Jacobi matrix has diagonal entries centered at V and off-diagonal entries controlled by $\sqrt{Vd}$. The exact finite inequalities are

$$|J_{ii} - V| \le 4\sqrt{Vd}, \qquad |J_{i,i-1}| + |J_{i,i+1}| \le 4\sqrt{Vd}.$$

Gershgorin gives the constant 8. The root/eigenvalue identification is the classical Jacobi matrix theorem [1]; the interval consequence is formalized as a typed adapter.

MSS Theorem 1.6 applied in original and reciprocal orientations produces $[u_-v_-, u_+v_+]$. Substituting the Jacobi intervals and the coarse $B/D \le 6$ yields

$$C_{\mathrm{loc}} = 12 + 8\sqrt{6} < 32.$$

The transposed expression $8 + 12\sqrt{6}$ is not used. The preliminary lower endpoint requires a constant strictly greater than 64; $K_{\mathrm{pre}} = 256$ is the chosen exact value.

10 Expanded recurrence checks

The terminating coefficient producer is initialized at $c_0 = 1$ and obeys

$$AC(k+1)(k+B+m)(k+D+m)c_{k+1} = D(k+m-d)(k+A+m)(k+C+m)c_k.$$

At $k = d-m$ the right side vanishes, proving termination. Acting on a monomial with $\vartheta = y\,\mathrm{d}/\,\mathrm{d}y$ multiplies it by its degree, so the Euler ODE is verified coefficientwise without an appeal to a symbolic ODE black box.

The derivative shift follows from the Pochhammer identity $(a)_{k+m} = (a)_m(a+m)_k$. Substitution in the ODE yields the four-term recurrence. The expanded coefficients are:

$$\begin{aligned}
P_{3,m} &= \frac{AC - Dy}{AC},\\
P_{2,m} &= B + D + 2m + 1 - \frac{Dy}{AC}(A + C + 3m - d + 3),\\
P_{1,m} &= (B+m)(D+m) - \frac{Dy}{AC}\big((A+m)(C+m)\\
&\qquad\qquad + (m-d)(A+C+2m+1)\big),\\
P_{0,m} &= -\frac{Dy}{AC}(m-d)(A+m)(C+m).
\end{aligned}$$

The last line is positive for $0 \le m < d$. Exact algebra proves equality of these expanded forms with the decomposed formulas used for estimates.

The property test evaluates 2355 rational identities across 27 legal tuples. The Mathematica clean-room test verifies all derivative orders at degrees $5, 8, 11$. These tests corroborate, but do not replace, the Lean coefficientwise proof.

11 Radius and residual constant propagation

The exact recurrence ledger fixes

$$C_0 = 48, \qquad C_1 < 96, \qquad K_r = 4096.$$

Each nonleading neighbor in the normalized recurrence is assigned a strict fraction of the global-maximum budget. Exact caps on d/n also ensure $B, D \ge K_0 d$ and

$$d + 2K_r\sqrt{Bd} \le \eta n.$$

Let $x_{\mathrm{admissible}}$ be the minimum of these rational caps. Since $\log(n+2) \le n$ for $n \ge 2$, the wedge implies $d/n \le K^{-1/3}$. Thus

$$K_{\mathrm{geometry}} = \lceil x_{\mathrm{admissible}}^{-3} \rceil + 1 = 52225595407351980347306803200000000000000001$$

forces every finite geometric inequality.

For the six-node residual,

$$|E_F(z)| \le \frac{C_{B6}}{720n^5 \log(n+2)} \prod_{j=0}^{5} |z-j|.$$

On the thickened domain each factor is bounded by 3ρ, and $\rho^6 \le K_r^6(3nd)^3$. The exponential step uses $|e^E - 1| \le 2|E|$ when $|E| \le 1$. All these factors form the exact symbolic $C_{\text{multiplier}}$ in the machine ledger. The two unresolved numerical inputs are only C_{B6} and N_{analytic}.

12 Lean theorem map and axiom audit

`XiMellin` through `ManuscriptCauchyTransport`. Concrete theta/Mellin/omega normalization, factor eight, quantitative fixed-sector saddle, leading contour, infinite higher-mode sum, Gamma/Stirling assembly, holomorphic coefficient theorem, and order-six Cauchy transport.

`HermiteGenocchiCube/FTC` and `ComplexHermiteGenocchi`. Six nested complex FTC steps, simplex mass, Newton identity, and local remainder. Boundary: the concrete $E_F^{(6)}$ bound.

`ElementaryCubeCalculus`. Fubini, shifts, $q = 1, \ldots, 4$, parameter derivatives, and errors. Boundary: xi-side uniform assembly.

`ExactXiBranch` through `XiNaturalSixCoefficientMatch`. Exact whole-box xi branch, inverse norm, positivity and ordering, transformed xi polynomial, and six coefficient matches at the explicit cutoff.

`TerminatingHypergeometric`. Finite coefficients, termination, derivative shift, Euler ODE, and recurrence. No finite-identity boundary.

`XiNaturalFiniteFreeSpecialization` through `XiNaturalCriticalRadius`.

Concrete finite-free specialization, including the coefficientwise bridge from the two displayed Jacobi factors to the published ${}_3F_2$, reciprocal roots, product arithmetic, complete critical-point localization, recurrence budgets, and normalized radius. Boundary: a factor-level typed MMP record, MSS, and classical Jacobi inputs; the final comparison's roots are not an assumed MMP field.

`XiNaturalConcreteMultiplier` through `XiNaturalMultiplierCertificate`.

The concrete xi multiplier, its six exact node values, its uniform unit bound, the finite Newton–Cauchy estimate, the canonical Rolle-point interval certificate, the equality with the actual transformed xi Jensen polynomial, and the headline negative-root conclusion are kernel checked. Boundary: the headline theorem takes only the narrow typed Jacobi, MMP, and MSS literature records already disclosed above.

The Phase-26 driver `Phase26Axioms.lean` prints all 1,217 audited declarations. Each depends only on the three standard Lean logical principles of propositional extensionality, classical choice, and quotient soundness. These are not project-specific mathematical axioms. Source scanners reject placeholders, custom axioms, and unsafe declarations on the formal proof surface.

The later Phase-28 driver separately prints the axioms of the three paper-facing terminal T5 declarations: the effective theorem, its literal existential R, C form, and the proportional-disc derivative statement through order six. Phase 29 isolates the same declarations in a Mathlib-only Palomar Challenge/Solution candidate and checks it locally with fresh-kernel replay, exact source-fidelity checks, and semantic mutations. Official Palomar Comparator and NanoDa replay is still pending; no official Palomar verification is claimed in this supplement.

Phase 30 extends the terminal audit through the xi-specific multiplier and assembly declarations. In particular, it audits the six-node theorem, the uniform unit bound, the concrete interval certificate, the transformed Jensen identity, and the headline theorem. Its remaining hypotheses are ordinary theorem parameters packaging the cited Jacobi, MMP, and MSS results; they are not project axioms.

13 Mathematica clean-room protocol

The Wolfram Language notebook was executed in Mathematica 15.0.1. It did not read JSON, Python output, SymPy output, Lean expressions, the frozen 82-term numerator, or any other packet artifact as mathematical input. The four groups were:

M1. Implicit differentiation of the saddle through order six, including the reduced H_5, H_6 denominator, term count, degree, and majorant.

M2. The terminating ${}_3F_2$ coefficient ratio, Euler ODE, four recurrence coefficients, and finite derivative-order tests.

M3. Positive solution of the limiting system, exact Jacobian, determinant, inverse, and infinity norm.

M4. Export of the exact result ledger and SHA-256 inventory.

The frozen hashes are listed in two 32-character blocks so that the values remain legible on a printed page:

Notebook. `C48_Mathematica_CleanRoom_2.nb`

`f79359ce48fe5e5015bceb88b5de82b87145da22e235688dfe3989526772dc7a`.

PDF. `C48_Mathematica_CleanRoom.pdf`

`f2e428bdd5d1caf03243b81dbc65794950abc2c8a4c13195395eca415d7a373c`.

Ledger. `C48_Mathematica_Result_Ledger.txt`

`1faea5fcb35b504b5d0aad9391999a99e530373dce36addb496eabb133fb04cc`.

The Phase-25 verifier recomputes these hashes, parses the exact ledger, rejects machine-real values in exact outputs, and scans the notebook for external input operations. The notebook was entered and run by the author from AI-prepared cells, so this is an independent CAS channel but not independent human review.

14 Arb/ACB directed verification

The directed suite uses `python-flint 0.6.0`, one thread, and fixed precision. The committed JSON is regenerated twice and compared byte for byte. The checks are:

1. Direct completed-zeta Taylor balls for $\gamma(0), \ldots, \gamma(3)$.
2. Positive omega integration and separate F-moment integration using six theta modes on $0 \le u \le 4$, analytic bounds for every omitted mode, and the complete $u > 4$ tail.
3. Strict Rouché inequalities, derivative nonvanishing, and $Q \neq 0$ on three small complex parameter boxes.
4. Directed contour, connector, central-window, and modes two and three at three sample points. This item is finite-grid regression only.
5. Implicit ACB power-series solution of the saddle through derivative order six without importing CAS formulas.
6. Isolated, pairwise-disjoint negative real root boxes for $(n, d) = (5, 3), (10, 4), (20, 5)$.

All infinite integral truncations have analytic tails. Conversely, no finite collection of contour samples is promoted to a uniform sector proof.

15 Source fidelity

The exact external statements consumed by the paper were checked against version-pinned sources.

1. MMP arXiv:2309.10970v3, not an unversioned preprint, supplies Proposition 2.7(iii), Definition 2.16, and Proposition 2.17. Version 3 is important because it corrects proposition text elsewhere in the paper.
2. MSS's published Theorem 1.6 is the maximum-root product inequality. Theorem 1.13 is a stronger S-transform statement and is not the theorem number cited here.
3. Holland arXiv:2608.08682v1 is used for architecture and formulas, not as a premise. The candidate does not import the assembled lemma whose $(C - D)/D$ hypothesis fails on the new branch.
4. The GORTTW v3 coefficient convention is compared to the direct xi normalization through the visible factor eight.

The publisher PDF for MMP was not openly available for byte comparison; the published bibliographic record is paired with the hash-frozen v3 arXiv text for exact proposition language.

16 Mutation and regression policy

Semantic mutations are designed to kill load-bearing changes, including:

- $n! \mapsto 2^{2n}$, $e^{u/2} \mapsto e^u$, and factor $8 \mapsto 4$;
- $r = 1/L$ or $\sigma = L/N$ reversal;
- fifth derivative substituted for sixth, or $1/720 \mapsto 1/120$;
- coordinate permutation in (α, t, w, δ);
- center-only replacement of a whole-box Jacobian quantifier;
- C_{loc} transposition, K_{pre} regression, or omission of the reciprocal MSS orientation;
- hypergeometric scale or recurrence sign changes;
- deletion of y^k or insertion of $1/k!$ in the radius;
- omission of C_{B6} or N_{analytic} from the final constant;
- any claim of human or peer review.

Equation regression also evaluates the xi coefficient identity and the critical-point radius at legal numerical points by an independent path. A string check alone is not accepted as evidence of a mathematical equation.

17 AI assistance and review disclosure

AI systems assisted with proof exploration, symbolic scripts, Lean code, manuscript drafting, source comparison, and adversarial review. Review packets were separated by analytic and algebraic scope, and later phases add separate Lean and reproducibility audits. Separation reduces some correlated failure modes but does not create a human reviewer. No report in the package is described as human review, expert certification, or peer review.

The AI systems used across the project include OpenAI Codex (GPT-5-family deployments), Anthropic Claude Opus 5, Moonshot AI Kimi K3, and Qwen3.8-Max via OpenRouter. Their roles included proof exploration, Lean and Python code generation, manuscript drafting, source comparison, and AI-only adversarial review; model-specific reports record their scope and context. AI systems are not authors. The author reviewed and edited the resulting material, accepts responsibility for the contents, executed the Mathematica cells, and authorized repository changes and release preparation. The exact-computation provenance is therefore different from a fully independent rederivation by another mathematician. The package is structured to make such a rederivation easier: expected values are not fed into the clean-room notebook, directed integral tails are explicit, and every formal boundary is named.

18 Reproduction entry points

From the repository root, the serial mathematical gates are

```
C48_PYTHON="$PWD/.venv/bin/python" \
  bash ground_zero_work/phase30/verify_phase30.sh
C48_PYTHON="$PWD/.venv/bin/python" \
  bash ground_zero_work/phase26/verify_phase26.sh
C48_PYTHON="$PWD/.venv/bin/python" \
  bash ground_zero_work/phase25/verify_phase25.sh
C48_PYTHON="$PWD/.venv/bin/python" \
  bash ground_zero_work/phase21/verify_phase21.sh
C48_PYTHON="$PWD/.venv/bin/python" ELAN_HOME="$HOME/.elan" \
  bash ground_zero_work/phase20/verify_phase20.sh
```

Do not run concurrent Lean builds, and do not run `lake update`. The Lean project is pinned by `lean-toolchain` and the Lake manifest. Python dependencies are pinned by the repository lockfiles. Generated Lake state, virtual environments, caches, and temporary files are not release sources.

The comprehensive package adds a top-level verifier with quick, full, and clean modes. The clean mode extracts a fresh source archive, verifies the manifest before execution, and rebuilds without inheriting local caches as logical evidence. Cached toolchain downloads may be used only as transport, not as unmanifested source.

References

[1] Gábor Szegő. *Orthogonal Polynomials*, volume 23 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 4 edition, 1975.